

Raghav Buddhineni

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EDUCATION

University of Michigan

Bachelor of Science in Engineering in Computer Science

GPA: 3.8/4.0 | **Awards:** William J. Branstrom Freshman Prize, Engineering Dean's List

Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Object-Oriented Programming

Organizations: Tech 4 Social Good (Project Manager), Michigan Ross Tech Plus Consulting (Tech Analyst)

Ann Arbor, MI

May 2028

EXPERIENCE

Machine Learning Researcher

UMTRI (University of Michigan Transportation Research Institute)

Ann Arbor, MI

Jun 2026 – Present

- Under **Professor Shan Bao**, engineered a multi-stage **Python data pipeline (Brave Search API, PRAW)** with keyword filtering to collect driver-reported ADAS distraction posts from 65+ automotive forums plus platforms like Reddit and X, running an automated discovery pass to build a **1,900+ record** corpus for downstream NLP analysis.
- Developed a custom filtering and preprocessing script to deduplicate at the complaint-ID level, clean, normalize, and enrich noisy user-generated data at scale, raising corpus quality for **BERTopic/LDA** topic modeling of driver distraction and automation complacency patterns; findings presented at the **2026 SURE Showcase**.

Tech Analyst

Michigan Ross Tech Plus Consulting

Ann Arbor, MI

Jan 2026 – Present

- Designed and shipped a **live, role-based operational dashboard** using **React** and **Firebase**, consolidating **10+** disparate logs into a **single KPI snapshot** with **15–20 metrics** and refresh cadences ranging from hourly to weekly.
- Piloted across **2 facilities** and completed a full client handoff with documentation, cutting manual reporting time by **30–40%** and enabling faster, data-driven operational decisions.

Co-Founder

WonderBox

Nashua, NH

Mar 2023 – Jun 2025

- Co-founded WonderBox, a 501(c)(3) nonprofit** combating loneliness among seniors in care homes through personalized care packages and handwritten letters.
- Led distribution of **100+** care boxes, built a team of **75+** volunteers as featured in local press, and spearheaded new chapter development nationwide.

Software Development Intern

Aurus Inc.

Norwood, MA

May 2024 – Aug 2024

- Installed OS, software, and **encryption keys** to conduct security protocols for **payment devices** used by **Fortune 500 retail clients**, ensuring compliance with deployment requirements.
- Validated device configurations** and **ensured compliance** with **customer SLAs**; coordinated technical tasks and project timelines across multiple client accounts.

PROJECTS/RESEARCH

Gradescope Study Tool (Chrome Extension) | *JavaScript, Firebase, Anthropic API*

Jan 2026 – Present

- Built a **Chrome extension** that scrapes student submissions and grading results from Gradescope, then calls the **Anthropic API** to generate a personalized diagnostic report identifying specific gaps in reasoning and conceptual understanding.
- Integrated **Firebase Authentication** and **Firestore** for secure user login and persistent storage of parsed grading data; connected to an external web platform leveraging the Anthropic API to deliver targeted practice questions. **Awarded Best Concept** at UMich MDC AI Hackathon.

Frat Map (Mobile App) | *Swift, Firebase, Twilio, Google Maps API*

Sep 2025 – Apr 2026

- Designed and developed a **cross-platform mobile social application** using **Swift**, implementing features including event discovery, friend grouping, and real-time coordination; built map-driven functionality with **Google Maps API** for live location sharing and proximity-based recommendations.
- Deployed a secure cloud backend using **Firebase Authentication** and **Firestore** with **Twilio-based** two-factor authentication (2FA), ensuring reliable data sync and user security; presented as one of **10** selected projects at V1 Demo Day.

Varroa Mite Detection Research | *Python, Pandas, TensorFlow, Keras*

Mar 2024 – Aug 2024

- Collaborated with **Stanford researchers** to build and train a **multi-layer convolutional neural network** using **TensorFlow** and **Keras** on bee image datasets, applying **transfer learning** and **data augmentation** to improve model generalization and detection accuracy.
- Processed and prepared large image datasets using **Python, Pandas**, and **NumPy**; utilized **Matplotlib** to visualize training progress, loss curves, and accuracy metrics across training epochs.

SKILLS/INTERESTS

Languages: Java, JavaScript/TypeScript, Python, C++, SQL, Swift, MATLAB, Julia

Technologies: React, Node.js, TensorFlow, Keras, HuggingFace Transformers, Pandas, NumPy, Matplotlib, BERTopic, Firebase, REST APIs, PostgreSQL/Supabase, Git, Excel

Interests: Playing Guitar, Soccer, NBA, Hair Care, Drinking Matcha, Traveling